**COEN 331 WIRELESS AND MOBILE NETWORKS**
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# RESEARCH ON BLUETOOTH
# (Technology, Protocol, Security)

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## *Audience*

The audience for this report is anyone who wants to learn about Bluetooth technology, protocol, and security. This report is specifically targeted to computer engineers, electronics engineers, product designers, mobile app developers, and anyone interested in wireless communication technologies. The audience is expected to have a basic understanding of electronics, networking and wireless communication concepts.

This report covers the overview of Bluetooth technology, including its architecture, Bluetooth with other IEEE technologies, and security. It also includes an in-depth discussion of Bluetooth protocol, including the various layers of the Bluetooth protocol stack and their functions. Along with that, it will cover the basic technical specs, operations and working in backend. The report also covers security issues in Bluetooth technology, including security modes, attacks and risk mitigation.

This report will discuss the most recent Bluetooth range extension and antenna, future of Bluetooth with respect to Drivers and Barriers, and its future scope of development in broad.

This report does not cover the basics of electronics, wireless communication concepts, or how to develop Bluetooth-enabled products or applications. It also does not provide specific instructions for using Bluetooth in various devices or systems.

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# 1. INTRODUCTION

Welcome to this report on Bluetooth Technology. In today's world, wireless connectivity has become an integral part of our daily lives. Bluetooth technology is one of the key players in the wireless communication arena. This report aims to provide a comprehensive understanding of Bluetooth technology, its architecture, operation, protocol stack, networking, interference issues, coverage, security, and future trends.

Chapter 1, the Introduction, will provide a brief overview of Bluetooth technology and its significance in the wireless communication domain. It will set the stage for the rest of the report, highlighting the key topics and objectives.

Chapter 2 will delve into the Architecture and Technical Working of Bluetooth technology, providing insights into its working mechanism. This chapter will cover the different components of Bluetooth technology and their functionalities.

Chapter 3, the Protocol Stack of Bluetooth, will discuss the various layers of the Bluetooth protocol stack, their functions, and how they interact with each other. Chapter 4 will cover Bluetooth Networking, focusing on the different networking topologies and their advantages and disadvantages.

Chapter 5 will address Bluetooth and IEEE 802.11 Interference Issues, explaining the potential interference between Bluetooth and other wireless technologies. Chapter 6 will cover Bluetooth Coverage, highlighting the limitations of Bluetooth technology in terms of range and coverage.

Chapter 7 will discuss Security and Interoperability, emphasizing the importance of security in Bluetooth technology and the measures taken to ensure interoperability between different devices. Chapter 8 will provide an in-depth analysis of the Security Model of Bluetooth, covering the various security features and mechanisms employed in Bluetooth technology.

Chapter 9, the Future of Bluetooth Technology, will focus on the potential applications and trends in Bluetooth technology. Finally, Chapter 10, the Conclusion, will summarize the key points of the report and provide recommendations for the future of Bluetooth technology.

Overall, this report aims to provide a comprehensive understanding of Bluetooth technology, its functionalities, and the future trends in the wireless communication domain.

# 2. BLUETOOTH TECHNOLOGY FUNDAMENTAALS

Bluetooth technology has revolutionized the way devices communicate wirelessly over short distances. It is a ubiquitous communication standard that has become an integral part of our daily lives, allowing us to connect and transfer data between a wide range of devices seamlessly. This chapter provides a detailed understanding of Bluetooth technology, exploring its concept, technical specifications, internal workings, and operation. By the end of this chapter, audience will have gained a comprehensive understanding of Bluetooth technology and its applications in modern technology.

## 2.1. What is Bluetooth Technology?

Bluetooth technology is a wireless communication technology that has transformed the way we connect and share information between devices. It enables devices to connect and interact with each other over short distances using radio waves. Bluetooth technology has become an integral part of our daily lives, and it is used in various devices, including smartphones, laptops, headphones, and smartwatches. The technology was first introduced in 1994 by Ericsson, and it has since undergone several updates and improvements to become a widely accepted standard for

wireless communication. Bluetooth technology has revolutionized the way we communicate, and it continues to be a crucial technology in the development of new devices and applications. [1] Below is the representation of Bluetooth connectivity with various different devices defined in a range in **figure 1**.

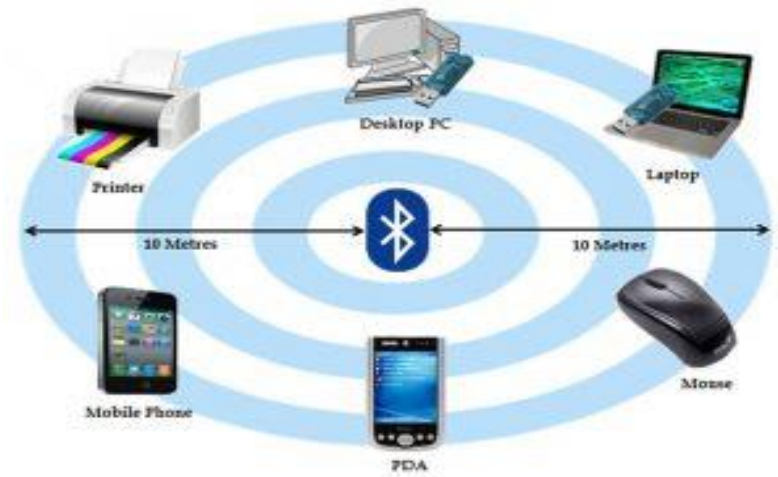


*Figure 1: Bluetooth connectivity with devices*

## 2.2. Technical Specification

Bluetooth is an universal for short range wireless voice and data communication. It is a Wireless Personal Area Network (WPAN) technology and is used for exchanging data over smaller distances. It operates in the unlicensed, industrial, scientific and medical (ISM) band from 2.4 GHz to 2.485 GHz. Maximum devices that can be connected at the same time are 7. Bluetooth ranges up to 10 meters. It provides data rates up to 1 Mbps or 3 Mbps depending upon the version. The spreading technique that it uses is FHSS (Frequency-hopping spread spectrum). A Bluetooth network is called a piconet and a collection of interconnected piconets is called scatternet.[2]

Like most wireless technologies, Bluetooth is fairly sophisticated and hence the complete technical specification is very vast. The **table 1** provides a summary of some key features of Bluetooth. [3]

| | |
|---|---|
| **Connection** | Spread Spectrum(Frequency hopping) |
| **Frequency band** | 2.4 GHZ ISM |
| **Modulation Technique** | Gaussian Frequency Shift Keying(GFSK) |
| **MAC Scheduling scheme** | FH-CDMA |
| **Transmission Power** | >20 dBm |
| **Aggregate Data Rate** | 0.721-1 Mbps |
| **Range** | 10m-100m |
| **Supported Stations** | 8 devices( per Piconet) |
| **Voice Channels** | 3 |
| **Data Security-Authentication key** | 128 bit key |
| **Data Security-Encryption key** | 8-128 bits(configurable) |

Table 1: Bluetooth Technical Summary

### 2.3. Working of Bluetooth

A Bluetooth Network is formed by a Personal Area Network, also known as a piconet, which comprises a minimum of two and a maximum of eight Bluetooth peer devices. Typically, there is a single master device and up to seven slave devices in a piconet. The master device is responsible for initiating communication with other devices and manages the communications link and traffic between itself and the slave devices associated with it. On the other hand, a slave device responds to the master device and must synchronize it is transmit/receive timing with that of the master device.[4]

The master device governs the transmissions of slave devices, dictating when they may transmit. A slave device can start transmitting only in a time slot immediately after being addressed by the master or in a time slot explicitly reserved for use by the slave device. The frequency hopping sequence is defined by the Bluetooth device address of the master device. The master device sends a radio signal to request a response from the specific slave devices within its range of addresses. The slaves respond and synchronize their hop frequency and clock with that of the master device.[4] If a device becomes an active member of more than one piconet, scatternets are formed. The device shares its time slots among the different piconets, allowing it to connect and communicate with devices in multiple piconets simultaneously.[4]

### 2.4. Bluetooth Operation

The concept of Bluetooth piconets and connection establishment are two fundamental components of the Bluetooth technology that enable the formation of wireless networks and data exchange between devices in close proximity. Let us discuss in brief the working of both the components of Bluetooth.[3]

1) Connection establishment
2) Concept of an ad-hoc piconet

Bluetooth networks are composed of nodes that are grouped together in piconets, which consist of a master node and up to seven slave nodes. A node may belong to multiple piconets and is referred to as a "bridge" in such cases. Communication in a piconet is half-duplex, with the master node being responsible for relaying messages between slaves. To communicate with nodes in other piconets, bridge nodes are necessary, but they cannot be active in multiple piconets simultaneously. Bluetooth nodes can be in different activity states, such as active, idle, parked, and sniffing, and data exchange occurs only when both nodes are active. The activity states of nodes change periodically. **Figure 2** provides an example of piconet organization [3].

The steps for pairing two Bluetooth devices are as follows:

1) Inquiry: This method entails a sender broadcasting inquiry packet that contain no other information except the sender's identify.

2) Page: A sender device attempts to establish a connection with a recipient device whose identification and clock are known while paging. For synchronization, page packets are exchanged, each of which includes the device address and clock of the sender.

# 3. ARCHITECTURE AND PROTOCOL STACK

This section of the research paper provides a detailed analysis of the architecture and protocol stack of Bluetooth technology, exploring the different layers and protocols involved and their roles in ensuring reliable and efficient communication between devices. By the end of this section, readers will have gained a comprehensive understanding

of the architecture and protocol stack of Bluetooth technology, providing a foundation for further research and development in this field.

### 3.1. Bluetooth Architecture

Bluetooth's architecture distinguishes between two categories of networks:

1) Piconet
2) Scatternet

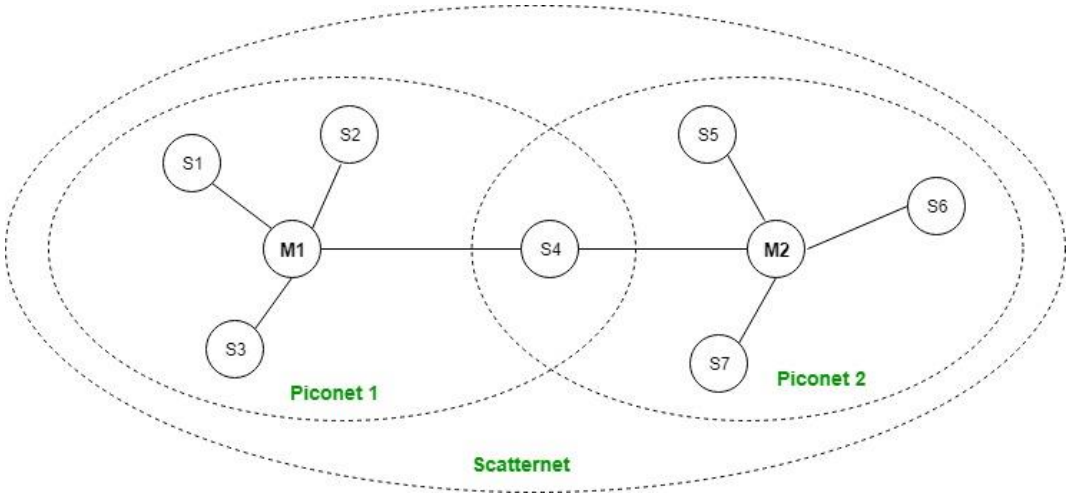


Figure 2: Bluetooth architecture

Piconet architecture:

One primary node, known as the master node, and seven active subsidiary nodes, known as slave nodes, make up the Piconet Bluetooth network type. As a result, we can state that there are 8 active nodes in total, all of which are located 10 meters apart. One-to-one or one-to-many communication between the primary and secondary nodes is possible. Slave-slave communication is not conceivable; only communication between the master and slave is potentially feasible. Also, it has 255 secondary nodes that are parked; these nodes cannot participate in communication unless they are activated.[2]

Scatternet architecture:

It is created by combining different piconets. A slave who is present in one piconet can serve as the primary or master in another. This type of node has the ability to receive a message from a master in one piconet and send it to its slave in the other piconet, where it is operating as a slave. A bridge node is the name for this kind of node. In two piconets, a station cannot be mastered.[2]

### 3.2. Protocol Stack

A protocol stack is a software- or hardware-based implementation of the actual protocols described in a standard that permits communication between devices based on that standard. **Figure 3** depicts the Bluetooth protocol stack in detail.

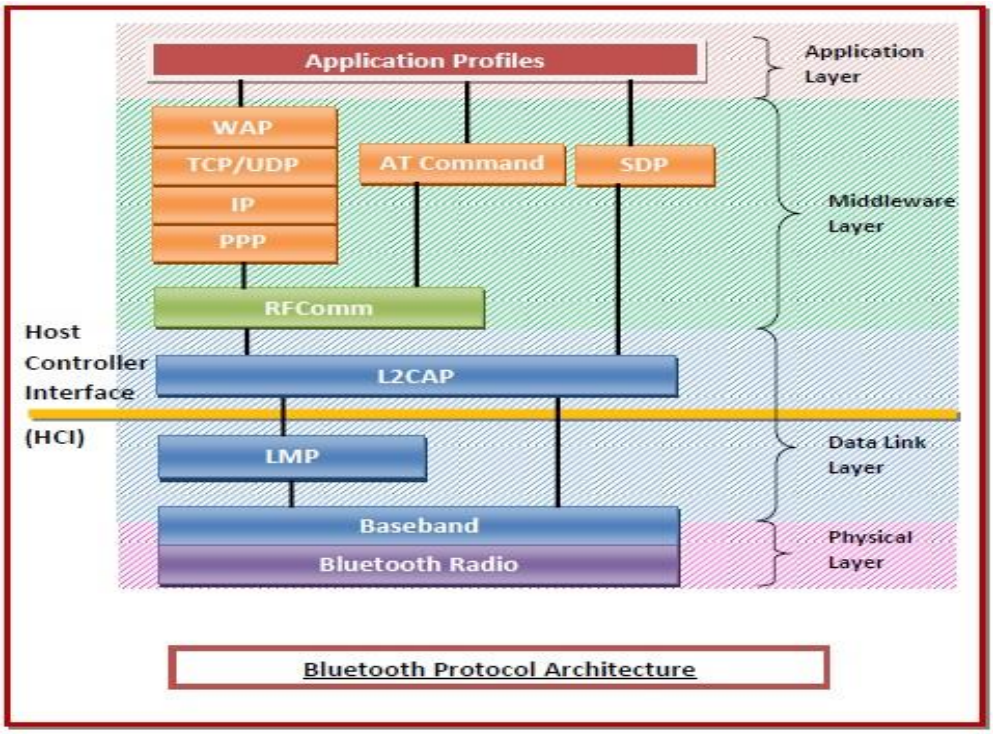


Figure 3 Bluetooth protocol stack

### *I. Radio Layer*

Radio layer, along with other wireless technologies, Bluetooth functions similarly. Bits (ones and zeros) are used to send data over a radio frequency. The radio layer specifies this function. Gaussian Frequency Shift Keying is used by Bluetooth transceivers (GFSK). Binary one is represented by a positive frequency deviation in GFSK, and binary zero is represented by a negative frequency deviation. There are three Power classifications for the Bluetooth transceivers. The range is clear of obstructions.[10]

| Class | Signal strength | Range |
|---|---|---|
| **Class 1** | 100 Millie watt | Up to 33 feet (10 meters) |
| **Class 2** | 2.5 mille watt | Up to 33 feet (10 meters) |
| **Class 3** | 1 mille watt | Up to 328 feet (100 meters) |

Table-2 Power classes [3]

Bluetooth uses the 2.4 GHz ISM (Industrial Scientific Medical) band of radio frequency. Most nations use the range of frequencies between 2400 and 2483.5 MHz. However, there are national restrictions on the frequency range in some nations. The band is divided into 79 MHZ channels (23 MHZ channels for Spain and France). Each of these channels is broken into 625 microseconds, with 1600 slots (1600 hops) being made every second. Due to frequency restrictions, there are specific frequency hopping algorithms for Spain, France, and Japan. Products using the limited band frequency (such as those sold in Spain) will not function with those using the full band (America). Hence, these goods must be viewed as local variations of a single.[6]

| Geography | Regulatory Range | RF Channels |
|---|---|---|
| USA, Europe and most other countries | 2.400-2.4835 GHZ | F=2.402+K(GHZ)<br>K=0,1,2……78 =(79) |
| Spain | 2.445-2.475 GHZ | F=2.445+K(GHZ)<br>K=0,1,2…….22=(23) |
| France | 2.4465-2.4835 GHZ | F=2.4465+K(GHZ)<br>K=0,1,2…….22=(23) |

Table-3 Frequency Spectrum

### *II. Baseband Link Layer:*

The baseband, which is analogous to the MAC sublayer in LANs, is the digital heart of a Bluetooth system. It handles addressing, packet format, timing, power control, and piconet connection setup.
The baseband layer creates two types of links that are determined by the application and operating environment. A Synchronous Connection Oriented (SCO) link mimics circuit switched connections for voice and data connections, while an Asynchronous Connection Link (ACL) is designed for data bursts. The ACL link can have symmetrical or asymmetrical data rates, which allows for broadcasting and data rate control by the master device. This makes it suitable for applications requiring Quality of Service (QoS).[10]

**Table 2**, also known as Power Classes, specifies the range of signal strength for each class. Class 1 has a signal strength of 100 Milliwatts and a range of up to 33 feet (10 meters), Class 2 has a signal strength of 2.5 milliwatts and a range of up to 33 feet (10 meters), and Class 3 has a signal strength of 1 milliwatt and a range of up to 328 feet (100 meters).[10]

A Bluetooth packet consists of three components: the access code, header, and payload. The access code is used for time synchronization, paging, and Piconet identification, and there are three types of access codes: Channel Access Code, Device Access Code, and Inquiry Code. The channel access code identifies a unique Piconet, while the device access code is used for paging and responses. The header contains information about packet acknowledgment, numbering, flow control, slave address, and error checking. The packet payload can contain data, voice fields, or a combination of both. The packet formats fall into nine categories: DM1, DH1, DM3, DH3, DM5, DH5, HV1, HV2, and HV3, each with its own specific purpose. Some packet types are designed for voice communication, while others are optimized for high-speed data transfer, with varying data rates. A Forward Error Correction scheme is used to minimize transmission errors for connections where Automatic Repeat Request (ARQ) is not available, and vice versa.[3][10]

### *III. Link protocol layer*

It manages links that have already been formed, including mechanisms for authentication and encryption. It is in charge of establishing the linkages, keeping an eye on their condition, and gracefully terminating them in response to instructions or errors. [2] The Logical Link Control and Adaptation Protocol (L2CAP) and the Link Management Protocol (LMP) are two protocols that allow for the construction of links and the accompanying control of such links. These protocols are necessary to control, among other things, encryption, QoS, and device pairing. [3][10]

### *IV. Service Discovery*

The service discovery protocol (SDP) identifies the services that are accessible within RF range and establishes their characteristics. Devices must be able to support the same services in order to connect to one another. SDP merely offers ways to find services; it does not offer ways to access them. This characteristic makes ad hoc networking possible, opening the door to a number of intriguing applications. It is also observed as one of Bluetooth's security problems is this particular attribute. [3][10]

***V. RFCOMM***

The RS 232 serial ports are simulated by the transport protocol known as RFCOMM. This standard supports functions like Bluetooth connections to printers and scanners. For dependable in-sequence transmission of byte streams, RFCOMM depends on the Bluetooth baseband. When a physical serial port is present, RFCOMM data rates will be constrained in the corresponding devices. Data pacing on virtual serial ports is an option for implementations. Framing, multiplexing, and the following other requirements are included in the straightforward, dependable RFCOMM transport protocol:[3][10]

Modem status - ring, RTS/CTS, DSR/DTR, DCD.

• Remote line status: Parity 17, overrun, and break
• Remote port parameters, including Baud rate, parity, and data bit count.
• Parameter bargaining (frame size)
• Optional flow control based on credit

The two categories of devices that RFCOMM supports.

Type 1: Internal emulated serial port

Type 2: Intermediate device with physical serial port[3][10]

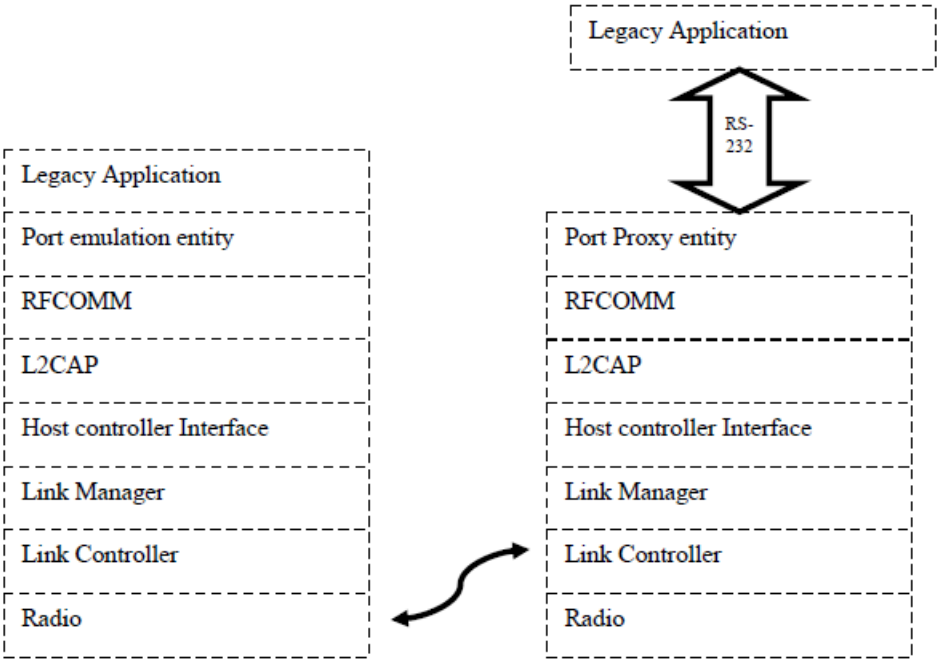


Figure-4: RFCOMM type 1 and type2 devices

### 3.3. Bluetooth and other wireless technologies

There are several wireless technologies which have various applications. Some of these technologies provide Bluetooth-like characteristics and applications. The **table-4** compares various wireless technologies based on different characteristics.

| Category | Home RF | Bluetooth | IEEE 802.11 | Wireless USB | IrDA | UWB |
|---|---|---|---|---|---|---|
| Purpose | Home | Cable/WPAN | WLAN | Cable | Cable | WPAN |
| Technology (2.4GHz) | FHSS | FHSS | FHSS/DSSS | DSSS | 850 nm | OFDM (3.1-10.6) |
| Power (dB/m) | 20 | 0-20 | 20 | 30 | --- | >>20 |
| Data Rate (Mbps) | 0.8/1.6 | 1 | 11 | 110/480 | 0.115/4 | 110-400 |
| Distance (meters) | 50 | 10-100 | 30-100 | 10< | 3/5 | 3/10 |
| Topology (Devices) | 128 | 8 | 128 | 127 | 10 | 127 |
| Security | Optional | Authentication + Encryption | Optional WFP | Application layer | Application layer | Low |

Table 4 Comparison of wireless technology[3]

We can see from comparing data from Table-4 that there are various wireless technologies with applications that overlap with Bluetooth. All of them have some restrictions and don't offer the same level of coverage as Bluetooth. We will cover them in detail below.

### *I. UWB*

Ultra Wideband Radio (UWB) is a novel wireless technology that employs a concept similar to radar. This technology is commonly used in Wireless Personal Area Networks (WPAN) and provides a data transfer rate of up to 400Mbps over a range of 30 feet while utilizing minimal power. In terms of capacity and power consumption, UWB outperforms Bluetooth. Furthermore, UWB has the ability to transmit multiple audio and video streams due to its ample bandwidth capacity. Unlike other radio frequencies, UWB does not cause any interference to other devices like broadband television sets, cell phones, or cordless phones. As a result, UWB poses a significant threat to Bluetooth.[3]

### *II. HOME RF Working Group*

Home RF and Wireless USB are two technologies that have some similarities in application with Bluetooth. However, it's worth noting that Home RF has not gained as much support in the industry as Bluetooth, mainly because it is limited to home networking due to its higher power consumption and lack of profiles. Therefore, it is not suitable for use in personal mobile devices, unlike Bluetooth. Similarly, Wireless USB is a technology that allows the emulation of a USB cable over wireless, but it does not have the same level of capability to support various profiles as Bluetooth does.[3]

### *III. WLAN*

WLAN (IEEE 802.11) appears to outperform Bluetooth in most aspects, but it is known for its high power consumption as it is based on the Ethernet standard. While WLAN is primarily used for local area networking, it may not be the best option for applications such as cable replacement or WPAN due to its high power consumption. There are also concerns about whether the high-end bandwidth can be maintained with IEEE802.11 technology, as the maximum data rates are up to 5 Mb/s. Additionally, the cost of these products starts from $100 and can go up to $200 for HR versions, which is significantly higher than that of Bluetooth devices.[3]

### *IV. Infrared Data Association*

IrDA is a cable replacement technology that is simple and efficient, but its use of optical transmission limits it to short-range line-of-sight communication. It also lacks networking features that Bluetooth offers, as it is a point-to-point communication technology. However, because of the line-of-sight communication, the chance of eavesdropping is remote, which makes IrDA inherently secure. Nevertheless, Bluetooth has enhanced security features that make up for this limitation. The IrDA standards, which were published in 1993, include the Serial Infrared (SIR) link specification, and in 1995, extensions were released to enable 4 Mb/s operations.[3][10]

| Parameter | Bluetooth | IrDA |
|---|---|---|
| Medium | RF/Omni-directional | Optical/Directional |
| Maximum Range | 10m/100m | 20cm/1.2m |
| Data and Voice | Data and Voice | Data only |
| GDR (Mb/s) | 1 | 1.52,4,16 |

Table-5 IrDA Standards [3]

## 3.4. Bluetooth and IEEE 802.11 Interference Issues

Bluetooth and IEEE 802.11 are two wireless technologies that operate in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band. While both technologies have their own unique advantages and applications, their coexistence in the same frequency band can cause interference issues that can impact their performance and reliability.[3]

The main cause of interference between Bluetooth and IEEE 802.11 is that they use the same frequency band and modulation scheme. This means that the radio signals from these technologies can overlap and interfere with each other, resulting in reduced throughput, packet loss, and increased latency. This interference can occur in both directions, where Bluetooth transmissions can interfere with IEEE 802.11 transmissions and vice versa. [3]

To address these interference issues, both Bluetooth and IEEE 802.11 have implemented various mechanisms and techniques. Bluetooth uses frequency hopping and adaptive frequency hopping techniques to minimize interference, while IEEE 802.11 uses carrier sensing and contention avoidance mechanisms to reduce collisions with other networks. Additionally, newer versions of both technologies, such as Bluetooth 5 and IEEE 802.11n, use wider bandwidths and advanced modulation techniques to improve their performance and reduce interference. [3]

Despite these measures, interference between Bluetooth and IEEE 802.11 can still occur in certain scenarios, such as in densely populated areas or in environments with many wireless devices. This can be a significant concern for applications that require high reliability and low latency, such as real-time audio and video streaming. As such, it is important for designers and engineers to carefully consider the potential interference issues when deploying wireless networks that use both Bluetooth and IEEE 802.11 technologies. [3]

In conclusion, while Bluetooth and IEEE 802.11 are both widely used wireless technologies that offer a range of applications and benefits, they can interfere with each other when operating in the same frequency band. It is important to carefully manage and mitigate these interference issues to ensure reliable and effective wireless communications. Ongoing research and development in this area will be critical to further improving the coexistence of these technologies and addressing the challenges of interference in the future. [3]

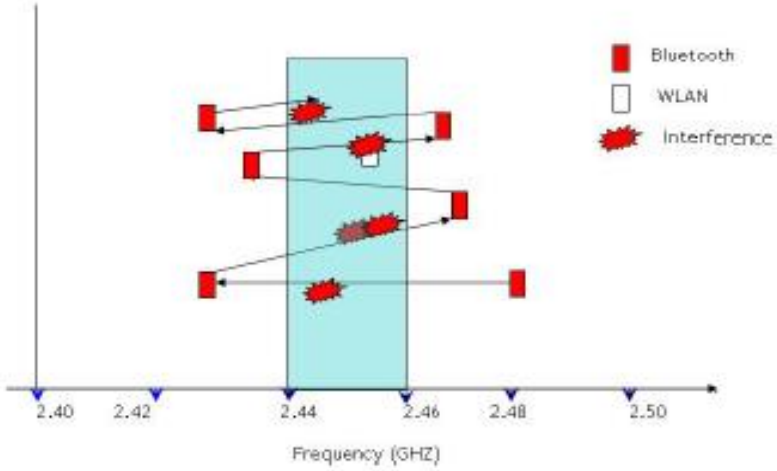


Figure-5: Bluetooth and WLAN interference

# 4. BLUETOOTH COVERAGE

Bluetooth coverage refers to the distance over which a Bluetooth device can communicate with another device. The coverage of Bluetooth devices is dependent on various factors such as the transmission power of the device, the receiver sensitivity, and the presence of obstacles between the devices. Bluetooth devices typically have a range of up to 10 meters, but this can be extended up to 100 meters in some cases, depending on the circumstances. However, the actual range can be affected by environmental factors such as walls, furniture, and other objects that can block or reflect the signal. In general, Bluetooth coverage is limited to within the same room or nearby areas, making it ideal for personal and local area network applications.

The range of Bluetooth depends on its class and there are three main classes of Bluetooth: [7]

Class 1 : transmits at 100 mW with a range of 100 meters.
Class 2 : Most Bluetooth headsets and headphones transmit at 2.5 mW with a range of 10 meters. Class 3 : transmits with 1 mW at a range of less than 10 meters.

## 4.1. Bluetooth Range Extension

Spatial constraints hamper communication when the devices are part of a larger, widely distributed network. Newer versions of Bluetooth have ranges from about 250 feet to 800 feet, so updating your device seems to be the most lucrative solution[7].

However, ranges can also be stretched by using signal repeaters, intermediate devices that capture, amplify, and then transmit or repeat signals without distortion. For example, a device with a range of 33 feet that is connected to a 1000 foot repeater can enjoy a range of 1,000 feet [7].

## 4.2. Bluetooth Antenna

Bluetooth antennas serve the purpose of extending the range of Bluetooth by transmitting and receiving signals. These antennas have various characteristics, including gain and direction, and their performance is critical as they are used

for both transmitting and receiving signals. However, there are certain issues related to Bluetooth antennas, such as power coupling losses between the antenna and the transmitter, as well as placement concerns.[3]

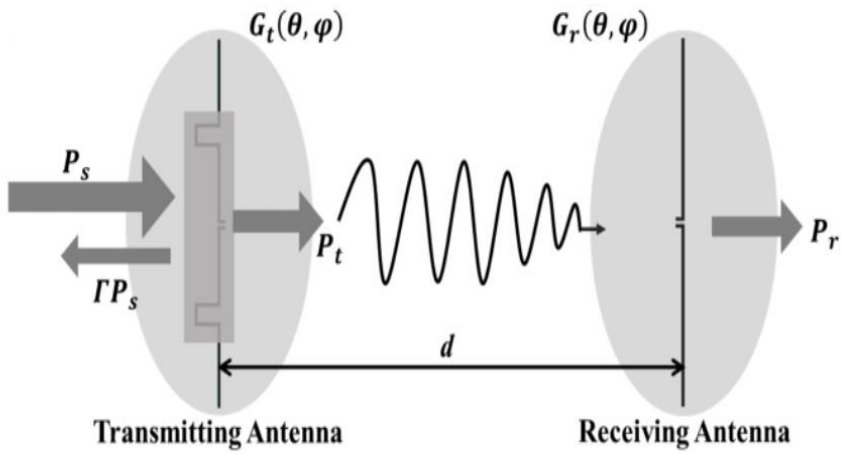


Figure 6: Send and receive blocks of Bluetooth antenna

The process and function of a Bluetooth antenna are illustrated in **Figure 5**. The antenna transmits the signal Pi to the receiver, while Pt is the transmitted signal. Additionally, Pr represents the signal reflected back to the antenna. The ratio of the transmitted power is denoted as Pi/Pt. It is important to match the maximum power radiated from the antenna with the impedance of the connected device, which is typically 50 ohms in most RF circuit cases. This is considered the optimal impedance for the antenna feed points.[3]

# 5. SECURITY

In the era of ubiquitous connectivity, Bluetooth has become a widely used technology for wireless communication between devices. While Bluetooth offers convenience and ease of use, it also presents potential security risks that cannot be ignored. With the increasing number of Bluetooth-enabled devices in use, the need for securing these devices from potential attacks has become more important than ever. Therefore, it is critical to understand the security challenges that Bluetooth technology presents and the best practices for securing Bluetooth devices. This section aims to provide an overview of Bluetooth security and the measures that can be taken to enhance the security of Bluetooth devices.

## 5.1. Bluetooth Attacks

*BlueSmacking:*

BlueSmacking is a type of Bluetooth attack that aims to disrupt the victim's mobile Bluetooth services. The attack involves sending a large number of big packets to the victim's device to create a service overload, resulting in a denial of service. In essence, BlueSmacking is a form of Distributed Denial of Service (DDoS) attack. BlueSmacking is often considered to be a less serious type of Bluetooth attack, as it does not directly result in data theft or privacy violations. However, the attack can cause a significant disruption in the victim's Bluetooth services, leading to the device being unable to connect to other Bluetooth devices. Additionally, the attack may decrease the battery life of the victim's device due to the excessive amount of data being transmitted. Despite being a relatively benign type of attack, BlueSmacking can still cause significant inconvenience and frustration to the victim.[8]

*Authentication Abuse*:

Authentication Abuse is a type of Bluetooth attack that targets all devices using a service on Bluetooth devices. When a device connects to the main device to use a service, it gains access to all other services running on the provider, including unauthorized services. The attacker leverages this vulnerability to connect to the unauthorized services and use them for their own purposes, enabling them to gain unauthorized access to the victim's device.[8]

*BlueDump Attack:*

The BlueDump attack requires the attacker to possess knowledge of the Bluetooth Device Address (BD_ADDR) associated with the victim's device. By replacing their own BD_ADDR with the victim's, the attacker can connect to the victim's device. However, as the attacker does not possess a connection key, the victim's device will reject the connection request and respond with a "HCI_Link_Key_Request_Negative_Reply". In some cases, this response may cause the victim's device to clear its connection key and enter pairing mode again.[8]

*BlueChop Attack:*

The BlueChop attack leverages the capability of a primary device to connect to several secondary devices, thereby forming an expanded network or Scatternet. The aim is to disrupt the network and break Pictonet connections for devices linked to Scatternet. The attacker accomplishes this by substituting their device's address with the address of a device that's already connected to Pictonet, then establishing a connection with the primary device. This action results in the disruption of the Pictonet connection. To carry out the BlueChop attack, the attacker may exploit the toggles in the Android Quick Settings menu, such as Bluetooth and Airplane mode.[8]

*BlueBump Social Engineering Attack:*

The BlueBump attack is a social engineering tactic that requires the attacker to establish a seemingly secure connection with the victim. This is often achieved by using a virtual job card or a file transfer. Once the victim adds the attacker's device to the trusted device list after the transfer, the attacker will request the victim to delete the connection key without severing the connection. The victim may not realize that the attacker is still connected, and continues to use their device as normal. The attacker will then request to re-key using the current connection, enabling their device to re-enter the victim's trusted list without proper authentication. Subsequently, the attacker can access the victim's device until the victim disables the key.[8]

## 5.2. Security Features and Modes

Four security modes are outlined in the various Bluetooth specifications. All four Bluetooth modes are supported by some Bluetooth versions, but not all. Each Bluetooth device must function in one of the four following modes: -

- Security Mode 1:
  It is insecure to use Security Mode 1. In essence, Bluetooth devices operating in this mode are "Promiscuous" and don't use any safeguards to prevent connections with other Bluetooth-enabled devices.[11]
- Security Mode 2:
  The centralized security manager oversees access control policies and communicates with users of different protocols and devices. Every Bluetooth gadget is capable of supporting Security mode 2. [11]
- Security Mode 3:

Before the physical link is entirely formed, a Bluetooth device starts security processes. All connections to and from Bluetooth devices that are in Security Mode 3 are required to be authenticated and encrypted. Both encryption and unidirectional or mutual authentication are supported in this mode. When the two devices initially communicate, a pairing mechanism is utilized to generate this key. When two Bluetooth devices are "attached," or in the process of interacting, the link key is generated. According to the Bluetooth protocol, when a user inputs the same PIN into two linked devices, they simultaneously generate link keys during the startup phase. Figure 6 conceptually represents the key derivation, device association, and PIN entry. Following initialization, devices operate automatically and transparently**.** Below **Figure-7** depicts the Bluetooth generation key from the PIN. [11]

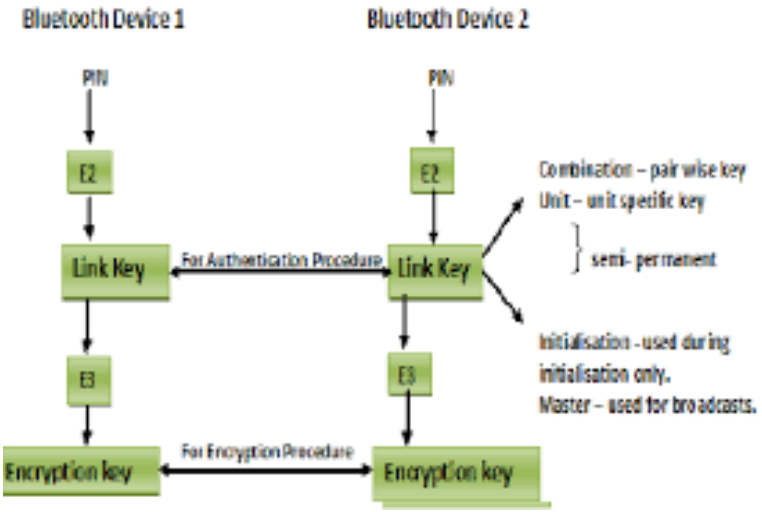


Figure 7: Bluetooth Generation Key from PIN

- Security Mode 4:

Similar to Security mode 2, this mode. A service level enforced security mode known as security mode 4 starts security procedures after link establishment. Secure Elliptic Curve Diffie Hellman (ECDH) methods are used in Simple Pairing for link key generation and key exchange. Services protected by Security Mode 4 must comply with one of the following security requirements: authenticated link key required, unauthenticated link key required, or no security required. [11]

### 5.3. Risk Mitigation and Counter Measure

To achieve risk mitigation in Bluetooth systems, countermeasures can be applied to address specific threats and vulnerabilities. It is crucial for organizations to provide an adequate level of knowledge and understanding to individuals who will be dealing with Bluetooth-enabled devices. Designing and documenting security policies that address the use of Bluetooth-enabled devices and user responsibilities, and including awareness-based education can also help enhance staff understanding and knowledge of Bluetooth.[9]

- Develop an organizational wireless security policy that addresses Bluetooth technology

- Ensure that Bluetooth users on the network are made aware of their security-related responsibilities regarding Bluetooth use
- Conduct comprehensive security assessments at regular intervals to fully understand the organization's Bluetooth security posture
- Ensure that wireless devices and networks involving Bluetooth technology are fully understood from an architecture perspective and documented accordingly
- Provide users with a list of precautionary measures to better protect handheld Bluetooth devices from theft
- Change the default setting of the Bluetooth device to reflect the organization's security policy
- Set Bluetooth devices to the lowest necessary and sufficient power level so that transmissions remain within the secure perimeter of the organization
- Choose PIN codes that are sufficiently random and long, avoiding static PINs such as all zeroes.[9]

# 6. FUTURE OF BLUETOOTH TECHNOLOGY

The object of the sections is to investigate the prospective advancements in Bluetooth technology, with a particular focus on its potential in driver and barrier applications. Moreover, this paper will review the latest developments that are likely to shape the future of Bluetooth, such as the emergence of Bluetooth Low Energy (BLE) and advancements in mesh networking. By examining these innovations, this paper aims to provide insights into how Bluetooth can continue to serve as an essential part of our interconnected world.

## 6.1. Drivers and Barriers

The Bluetooth Special Interest Group (SIG) is identified as a strong driver for the technology, as it ensures that any product labeled as Bluetooth-enabled conforms to their specification and is seamlessly interoperable with other manufacturers' Bluetooth-enabled devices. The availability of Bluetooth in mobile phones, coupled with the availability of several profiles in the specification, paved the way for deeper penetration of Bluetooth in the market.

Bluetooth in portable computers, personal digital assistants, and peripheral devices like printers and keyboards. The increasing functionality of mobile phones, including new applications such as internet browsing, multimedia, gaming, and m-commerce applications, has fueled the demand for Bluetooth as it offers easy wireless connectivity without the need for cables.[3]

One of the barriers to the adoption of Bluetooth technology is the slow vendor adaptation, where vendors are slow in adapting the technology, leading to interoperability issues. Although the Bluetooth SIG requires that products of different vendors be interoperable, this has not always been the case due to various reasons, including some far east vendors supplying Bluetooth devices without passing through the qualification process. Another barrier is the emergence of wireless USB, which offers a simpler solution to replace USB cables, posing a threat to Bluetooth. However, vendors like Nokia believe that wireless USB is a complementary technology and not a real threat or barrier to Bluetooth.[3]

## 6.2. Future Development

Bluetooth's success can be attributed to the Bluetooth Special Interest Group (SIG), which develops and upgrades the standard. Bluetooth's latest version, 3.0 with enhanced data rate, increases data rate from 24Mbps to 3Mbps and aims to reduce power consumption by 50%. The SIG has decided to collaborate with the Ultra Wide Band (UWB) work group to incorporate UWB radio technology into Bluetooth, which offers much higher data rates of up to 100Mbps, making it a smart move to leverage both the branding and technical advantages of Bluetooth and UWB.

Future applications of Bluetooth can be divided into different categories. Firstly, the cost reduction of Bluetooth chipsets, along with a decrease in power consumption, will enable their use in disposable sensors and personal tags, such as in the pilot project in Denmark, where Bluetooth-enabled tags are given to children at the zoo for their safety. Secondly, mobile phones, which are increasingly becoming personal information management devices, can use Bluetooth to carry electronic identity, which can be used for authentication and authorization, purchase of goods, and enhancing security in offices. Lastly, Bluetooth can be a simple and reliable technology for location determination, which can be used to push personalized services, advertisements, and information to users' mobile phones in shopping malls, fun-fairs, and other locations.

# 7. CONCLUSION

In conclusion, Bluetooth technology has come a long way since its inception in 1994, and it has revolutionized the way we interact with technology. It has become a standard for short-range wireless communication, enabling seamless connectivity between various devices. The protocol

has undergone several upgrades, with the latest version providing enhanced data rate and reduced power consumption. However, like any other technology, Bluetooth is not without its security risks. Several vulnerabilities have been discovered over the years, and while the Bluetooth SIG and other stakeholders have taken steps to address them, more work needs to be done to ensure the security of Bluetooth devices and networks.

In recent years, research has focused on identifying and addressing security issues in Bluetooth technology, and several solutions have been proposed, such as the use of encryption, authentication protocols, and intrusion detection systems. As the use of Bluetooth continues to expand into new areas such as the Internet of Things (IoT), it is essential that security concerns are taken into account to prevent potential attacks. Looking into the future, the collaboration between the Bluetooth SIG and the UWB workgroup to incorporate UWB technology into Bluetooth is a promising development. This will enable higher data rates and improve the performance of Bluetooth devices, further expanding the potential for new applications and use cases.

In summary, Bluetooth technology has proven to be a game-changer in the world of wireless communication, and its impact will continue to be felt for many years to come. With continued research and development, coupled with the implementation of robust security measures, Bluetooth technology will remain a critical enabler of connectivity and convenience in our increasingly connected world.

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# ACRONYMS

| | |
|---|---|
| **BLE** | Bluetooth Low Energy |
| **IEEE** | Institute of Electrical and Electronics Engineers |
| **WPAN** | Wireless Personal Area Network |
| **ISM** | Identity and Security Management |
| **FFHSS** | Frequency-hopping spread spectrum |
| **FH-CDMA** | Frequency Hopping Code Division Multiple Access |
| **MAC** | Media Access Control |
| **MHZ** | Mega Hertz |
| **ACL** | Asynchronous Connection Link |
| **ARQ** | Automatic Repeat Request |
| **RF** | Radio frequency |
| **SDP** | Service discovery protocol |
| **RFCOMM** | Radio frequency communication |
| **RTS** | Request To Send |
| **CTS** | Clear To Send |
| **DSR** | Demand signal repository |
| **UWB** | Ultra Wideband Radio |
| **WLAN** | Wireless Local Area Network |
| **HR** | High rate |
| **SIR** | Serial Infrared |
| **ISM** | Industrial, Scientific, and Medical |
| **BD_ADDR** | Bluetooth Device Address |
| **PIN** | Personal identification number |
| **ECDH** | Elliptic Curve Diffie Hellman |
| **SIG** | Special Interest Group |